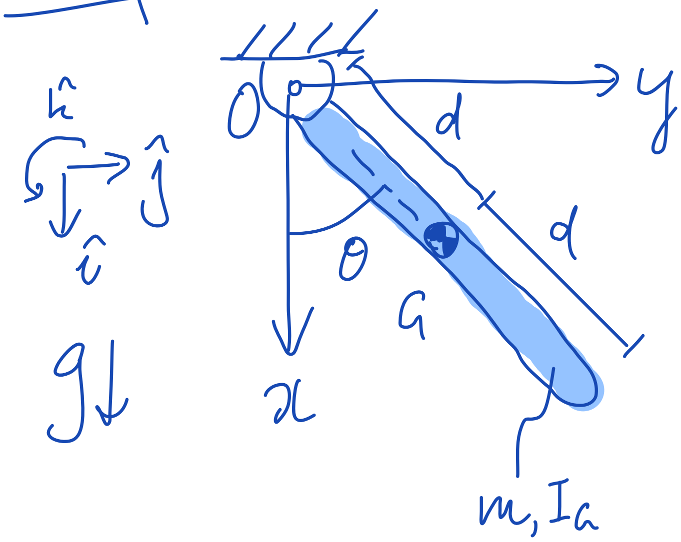
Compound Pendulum

- Minimal Coords {3 methods}
- Maximal Coords

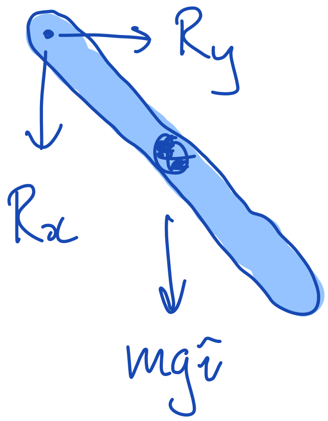
Prelim 1.
Individual Canvas "Quiz" → don't click til you are ready.

Setup

Find EoM



FBD



Minimal Coords

Find $\ddot{\theta}$ in terms of $\dot{\theta}, \theta$

Need to choose N-E equations.

AMB/O $\qquad \sum \vec{M}_{ext/O} = \dot{\vec{H}}_{/O}$

↳ Eliminates R_x, R_y from analysis.

1st Method

$$\vec{H}_{/O} = I_O \dot{\theta} \hat{k}$$

$$= (I_G + md^2)\dot{\theta}\hat{k}$$

Parallel Axis Theorem.

$$\dot{\vec{H}}_{/O} = (I_G + md^2)\ddot{\theta}\hat{k}$$

True because O is a fixed point.

Sub into AMB

$$\sum \vec{M}_{ext/O} = \dot{\vec{H}}_{/O}$$

$$\underbrace{d\sin\theta\hat{j}}_{\vec{r}} \times \underbrace{mg\hat{i}}_{\vec{F}} = (I_G + md^2)\ddot{\theta}\hat{k}$$

$$\hat{k} \Rightarrow \quad -dmg\sin\theta = (I_G + md^2)\ddot{\theta}\hat{k}$$

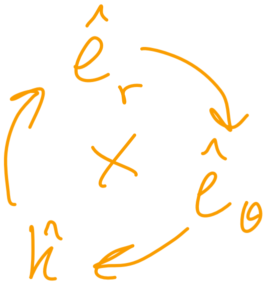
Soln

$$\boxed{\omega = \dot{\theta}, \quad \dot{\omega} = \ddot{\theta} = -\frac{dmg\sin\theta}{I_G + md^2}}$$

Second Method $\dot{\vec{H}}_{/O}$

$$\boxed{\begin{aligned} a &= v^2/r \\ &= (r\omega)^2/r \\ &= r\omega^2 \end{aligned}}$$

$$\dot{\vec{H}}_{/O} = \dot{\vec{H}}_{G/O} + \dot{\vec{H}}_G$$



$$= \vec{r}_{G/O} \times m\vec{a} + I_G \ddot{\theta}\, \hat{k}$$

$$= d\hat{e}_r \times \left[m\left(\cancelto{0}{-d\dot{\theta}^2\hat{e}_r} + d\ddot{\theta}\hat{e}_\theta\right)\right] + I_G\ddot{\theta}\,\hat{k}$$

$$= md^2\ddot{\theta}\,\hat{k} + I_G\ddot{\theta}\,\hat{k}$$

$$= \left(I_G + md^2\right)\ddot{\theta}\,\hat{k}$$

Third Method

$$\dot{\vec{H}}_{/O} = \frac{d}{dt}\left(\vec{H}_{/O}\right)$$

$$= \frac{d}{dt}\int \vec{r} \times \vec{v}\, dm$$

$$= \frac{d}{dt}\int \vec{r} \times \vec{\omega} \times \vec{r}\, dm$$

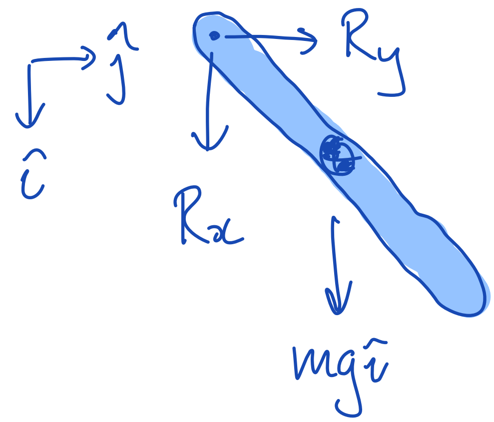
$$= \frac{d}{dt}\int_0^{2d} r^2\, dm\, \omega\, \hat{k}$$

$$= I_O\, \dot{\omega}\, \hat{k}$$

Maximal Coordinates Approach

Object is 'free' in space.

FBD



$\therefore$ maximal coords

x_a, y_a, θ

two unknowns R_x, R_y.

First NE then constraints.

LMB

$$\sum \vec{F} = m\vec{a}$$

$$R_x\hat{i} + mg\hat{i} + R_y\hat{j} = m(\ddot{x}\hat{i} + \ddot{y}\hat{j}) \quad (1,2)$$

AMB

$$\sum \vec{M}_a = \dot{\vec{H}}_{/a}$$

$$-d\sin\theta\hat{j} \times R_x\hat{i} - d\cos\theta\hat{i} \times R_y\hat{j} = I_a\ddot{\theta}\hat{k}$$

$$\hat{k} \Rightarrow \quad R_x d\sin\theta - R_y d\cos\theta = I_a\ddot{\theta} \quad (3)$$

Constraint

$$\vec{r}_o = \vec{r}_{o/a} + \vec{r}_a$$

$$0 = -d\cos\theta\,\hat{i} - d\sin\theta\,\hat{j} + x\hat{i} + y\hat{j}$$

$$\begin{cases} \hat{i} \Rightarrow & d\cos\theta = x \\ \hat{j} \Rightarrow & d\sin\theta = y \end{cases}$$

$$\frac{d^2}{dt^2} \begin{cases} -d\ddot{\theta}\sin\theta - d\dot{\theta}^2\cos\theta = \ddot{x} & (4) \\ d\ddot{\theta}\cos\theta - d\dot{\theta}^2\sin\theta = \ddot{y} & (5) \end{cases}$$

Soln

$$\begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{\theta} \\ R_x \\ R_y \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix} \begin{array}{c} (1) \\ (2) \\ (3) \\ (4) \\ (5) \end{array}$$